

Eenheid 6.4

Note Title

2015/03/26

Die kruisproduct p 808

Die kruisproduct word mbv "determinant" bepaal.
Definisie: Laat $a, b, c, d \in \mathbb{R}$

in Tweede orde determinant is

$$* \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Vbde: ① $\begin{vmatrix} -1 & 2 \\ 3 & 4 \end{vmatrix} = -4 - 6 = -10$

② $\begin{vmatrix} 3 & 4 \\ -1 & 2 \end{vmatrix} = 6 - (-4) = 6 + 4 = 10$

Definitie van kruisproduct. ($\vec{a} \times \vec{b} = \text{vektor}$) ant

Laat $\vec{a} = \langle a_1, a_2, a_3 \rangle$ en $\vec{b} = \langle b_1, b_2, b_3 \rangle$

$$\text{dan is } \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$
$$= (a_2 b_3 - a_3 b_2) \vec{i} - (a_1 b_3 - a_3 b_1) \vec{j} + (a_1 b_2 - a_2 b_1) \vec{k}$$

Opmerking: Laat $\vec{a}$ en $\vec{b}$ niet-nul vectoren in $\mathbb{R}^3$

wees. Die kruisproduct $\vec{a} \times \vec{b}$ ($\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$)
geeft een vektor wat loodrecht is op beide $\vec{a}$ en $\vec{b}$

$$\text{Dus } \begin{cases} (\vec{a} \times \vec{b}) \cdot \vec{a} = 0 \\ (\vec{a} \times \vec{b}) \cdot \vec{b} = 0 \end{cases}$$

VBD Bepaal $\bar{u} \times \bar{v}$ as $\bar{u} = \langle -3, 4, -2 \rangle$ en $\bar{v} = \langle 2, -1, 3 \rangle$

en toon aan dat jou antwoord korrek is

$$\bar{u} \times \bar{v} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -3 & 4 & -2 \\ 2 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix} \bar{i} - \begin{vmatrix} -3 & -2 \\ 2 & 3 \end{vmatrix} \bar{j} + \begin{vmatrix} -3 & 4 \\ 2 & -1 \end{vmatrix} \bar{k}$$

$$= (12 - 2) \bar{i} - (-9 - (-4)) \bar{j} + (3 - 8) \bar{k}$$

$$= 10 \bar{i} + 5 \bar{j} - 5 \bar{k} = \langle 10, 5, -5 \rangle$$

Toets: $(\bar{u} \times \bar{v}) \cdot \bar{u} = \langle 10, 5, -5 \rangle \cdot \langle -3, 4, -2 \rangle$

$$= -30 + 20 + 10 = 0$$

$$(\bar{u} \times \bar{v}) \cdot \bar{v} = \langle 10, 5, -5 \rangle \cdot \langle 2, -1, 3 \rangle$$

$$= 20 - 5 - 15 = 0$$

$$\bar{v} \times \bar{u} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & -1 & 3 \\ -3 & 4 & -2 \end{vmatrix}$$

Bepaal $\bar{v} \times \bar{u}$

$$= \langle -10, -5, 5 \rangle$$

Dus $\bar{u} \times \bar{v} = -(\bar{v} \times \bar{u})$

Oplossing

• Laat $\vec{a}, \vec{b} \in \mathbb{R}^3$, $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$, dan

1) $\vec{a} \times \vec{b}$ is loodreg / ortogonaal op beide $\vec{a}$ en $\vec{b}$

2) $\vec{a}$ en $\vec{b}$ is parallel aa $\vec{a} \times \vec{b} = \vec{0} = \langle 0, 0, 0 \rangle$

3) $|\vec{a} \times \vec{b}|$ is die oppervlakte van die parallelogram wat deur $\vec{a}$ en $\vec{b}$ bepaal word.

Eier-skippe:

Laat $\bar{a}$, $\bar{b}$ en $\bar{c}$ vektore in $\mathbb{R}^3$ wees en $k \in \mathbb{R}$ ^{skalaar}

$$1. \bar{a} \times \bar{b} = -(\bar{b} \times \bar{a})$$

$$2. (k\bar{a}) \times \bar{b} = k(\bar{a} \times \bar{b})$$

$$\bar{a} \times (k\bar{b}) = k(\bar{a} \times \bar{b})$$

$$3. \bar{a} \times (\bar{b} + \bar{c}) = \bar{a} \times \bar{b} + \bar{a} \times \bar{c}$$

$$4. (\bar{a} + \bar{b}) \times \bar{c} = \bar{a} \times \bar{c} + \bar{b} \times \bar{c}$$

Voorbeeld: ① Laat $\bar{a} = \langle 1, 2, 0 \rangle$

$$\bar{b} = \langle 3, -1, 2 \rangle$$

$$\bar{a} \times \bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 2 & 0 \\ 3 & -1 & 2 \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ -1 & 2 \end{vmatrix} \bar{i} - \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} \bar{j} + \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} \bar{k}$$
$$= 4\bar{i} - 2\bar{j} - 7\bar{k}$$

$$= \langle 4, -2, -7 \rangle$$

Toets: $(\bar{a} \times \bar{b}) \cdot \bar{a} = 4 - 4 = 0$

$$(\bar{a} \times \bar{b}) \cdot \bar{b} = 12 + 2 - 14 = 0$$

② Laat $\vec{a} = \langle 2, 3, -1 \rangle$ en $\vec{b} = -2\vec{a} = \langle -4, -6, 2 \rangle$

* as $\vec{b} = c\vec{a} \Rightarrow \vec{a}$ en $\vec{b}$ is parallel

$\therefore \vec{a} \times \vec{b} = \vec{0}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & -1 \\ -4 & -6 & 2 \end{vmatrix} = (6 - (-6))\vec{i} - (4 - 4)\vec{j} + (-12 + 12)\vec{k} \\ = \langle 0, 0, 0 \rangle = \vec{0}$$

③ Bepaal in vektor met lengte 2 wat $\perp$ is op beide $\vec{a} = \langle 1, -1, 1 \rangle$ en $\vec{b} = \langle 3, 2, -1 \rangle$

- $\vec{a} \times \vec{b}$

- eenheidsvektor

- lengte 2 : $2 \left(\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \right)$

$$\bar{a} \times \bar{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 3 & 2 & -1 \end{vmatrix} = \quad \quad \quad = \langle -1, 4, 5 \rangle$$

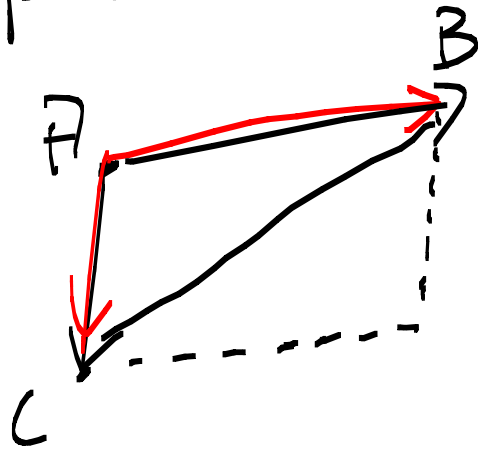
Toets : $(\bar{a} \times \bar{b}) \cdot \bar{a} = 0$
 $(\bar{a} \times \bar{b}) \cdot \bar{b} = 0$

Eenheidsvector : $\frac{\bar{a} \times \bar{b}}{|\bar{a} \times \bar{b}|}$

$$|\bar{a} \times \bar{b}| = \sqrt{(-1)^2 + (4)^2 + (5)^2} \\ = \sqrt{42}$$

vector met lengte 2 : $\frac{2 \times \langle -1, 4, 5 \rangle}{\sqrt{42}}$
 $= \frac{2}{\sqrt{42}} \langle -1, 4, 5 \rangle$

④ Bepaal de oppervlakte van die driehoek met
toppunten: $A(2, -3, 4)$ $B(1, 2, 3)$ $C(0, -2, -1)$



$$\Delta ABC = \frac{1}{2} \parallel \text{gram}$$

en opp van $\parallel \text{gram}$ is
 $|\overrightarrow{AB} \times \overrightarrow{AC}|$

$$\overrightarrow{AB} = \langle 1, 2, 3 \rangle - \langle 2, -3, 4 \rangle = \langle -1, 5, -1 \rangle$$

$$\overrightarrow{AC} = \langle 0, -2, -1 \rangle - \langle 2, -3, 4 \rangle = \langle -2, 1, -5 \rangle$$

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -1 & 5 & -1 \\ -2 & 1 & -5 \end{vmatrix} = (-25+1)\bar{i} - (5-2)\bar{j} + (-1+10)\bar{k} \\ &= -24\bar{i} - 3\bar{j} + 9\bar{k} \\ &= \langle -24, -3, 9 \rangle \end{aligned}$$

$$\begin{aligned}
 \text{Opp van 11 gram} &= |\overrightarrow{AB} \times \overrightarrow{AC}| \\
 &= \sqrt{(-24)^2 + (-3)^2 + (9)^2} \\
 &= \sqrt{576 + 9 + 81}
 \end{aligned}$$

$$\begin{aligned}
 \text{Opp van } \Delta &= \frac{1}{2} \text{ van opp van 11 gram} \\
 &= \frac{1}{2} \sqrt{576 + 9 + 81}
 \end{aligned}$$